

Chapter One

Overview of the internet and WWW

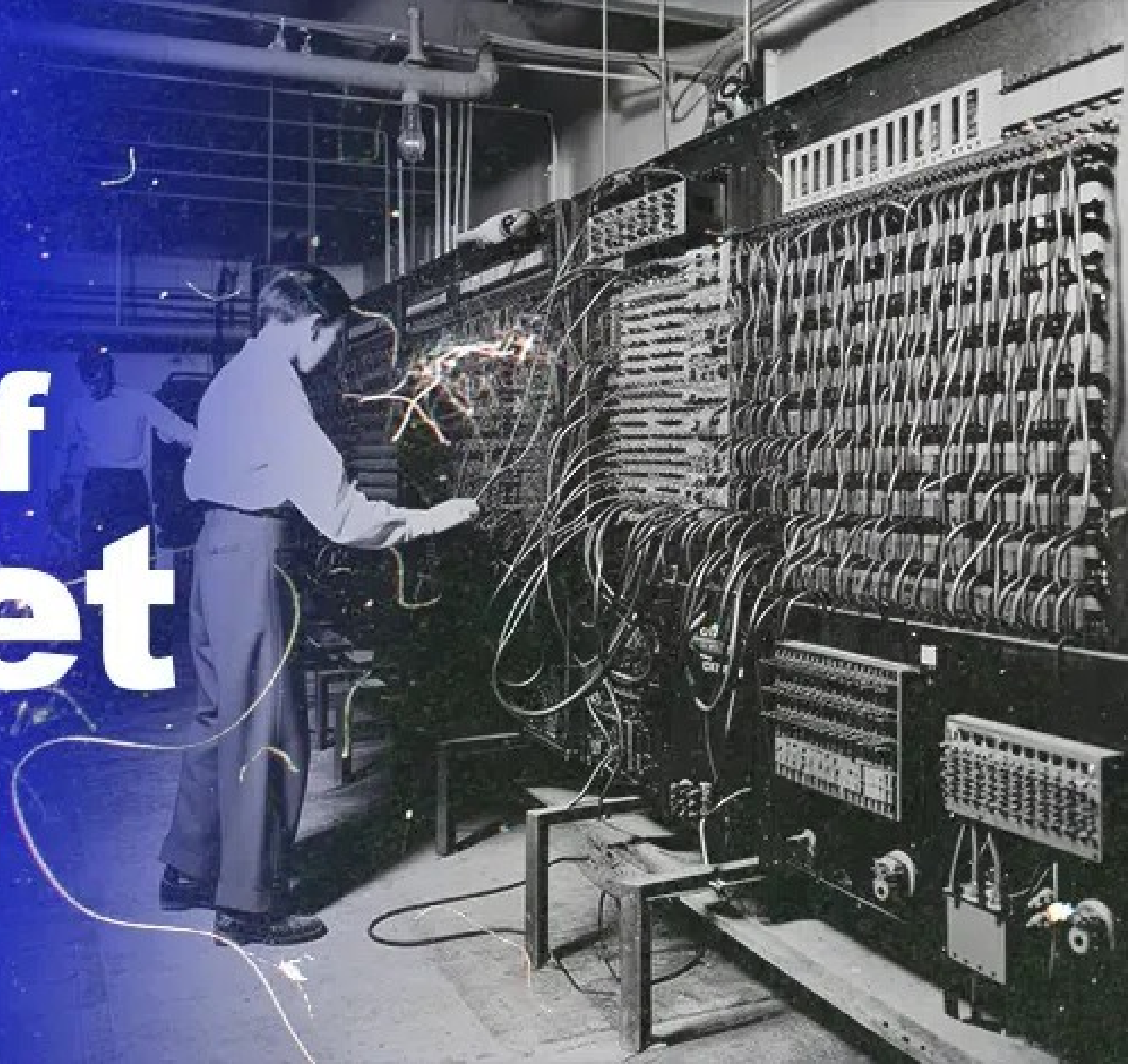


Outline

- Internet History, uses, and services
- Hypertext and the World Wide Web
- Introduction to web programming
- Programming vs. Scripting
 - Difference between programming and scripting
 - Commonly used scripting languages in web development
- Client/Server Model
- Universal Addressing
- Universal Protocols
- Website evaluation
 - readability, layout, coloring, trust...
- Basic UI/UX Design and concept

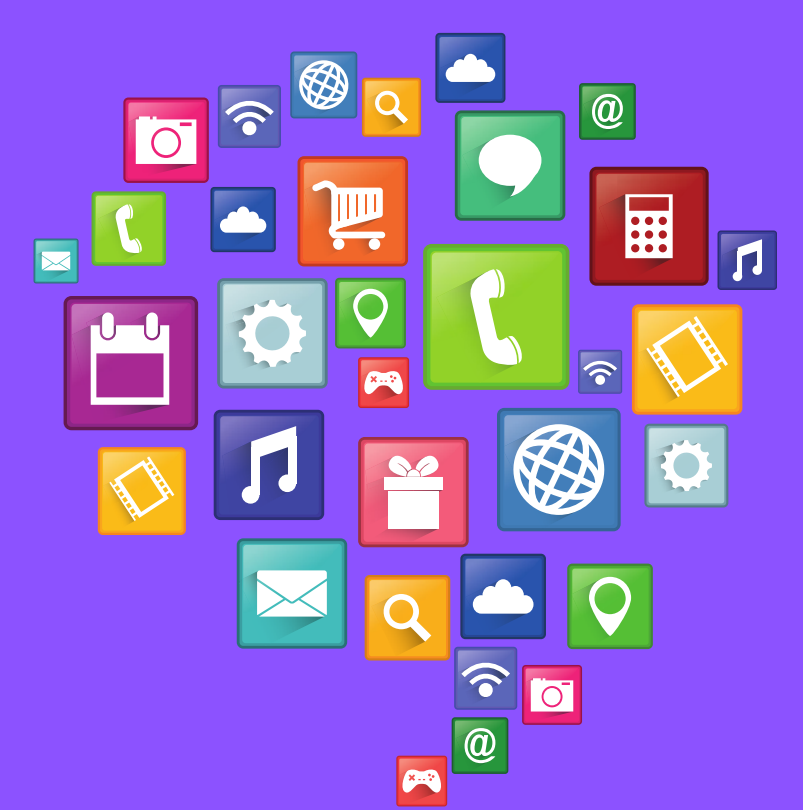


History of Internet



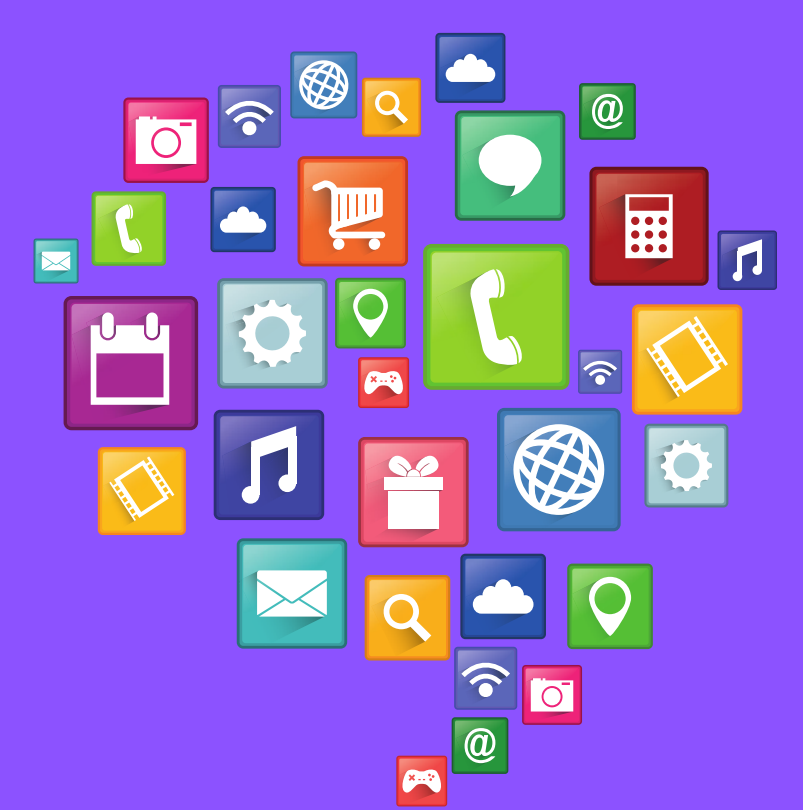
Internet History

- The internet as we know it today is a vast, interconnected network that has transformed how we communicate, work, learn, and interact.
- The origin of the internet dates back to the 1950s when the world was grappling with the Cold War, and researchers sought to create a communication system that could withstand disruptions.
- As cold war tensions grew and Sputnik was launched, the United States Department of Defense (DoD) began to seek additional methods of transmitting information to supplement existing methods. Sputnik was the world's first artificial Earth satellite, launched by the Soviet Union on October 4, 1957



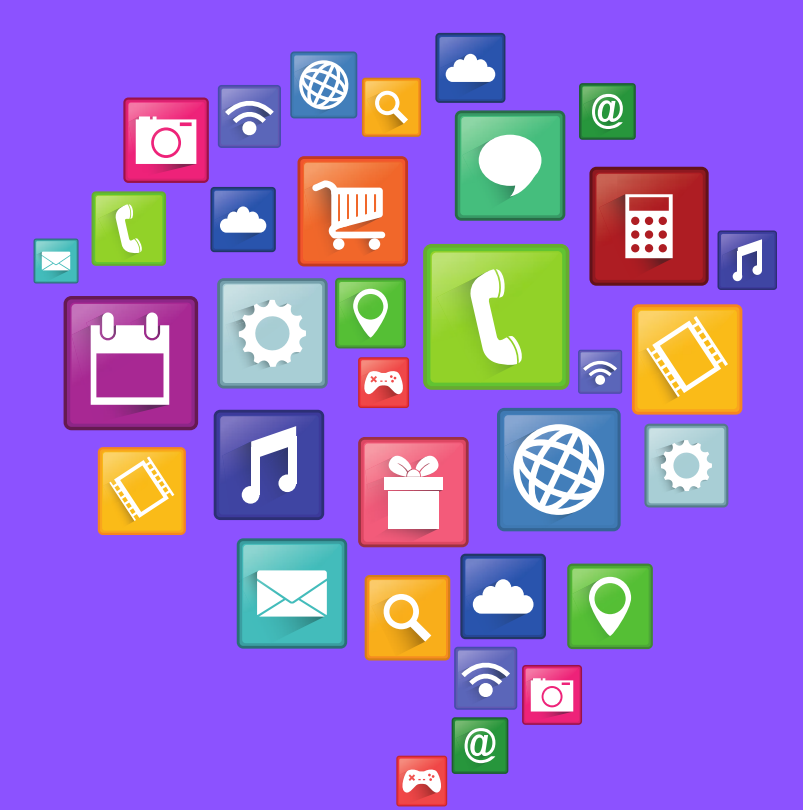
Internet History

- They sought something that was decentralized, allowing better resiliency in case of attack, where damage at one point would not necessarily disrupt communication.
- Their network, Arpanet(Advanced Research Projects Agency Network), connected the DoD and participating universities together for the first time.
- In order to standardize the way networked systems communicated the Transfer Control Protocol/Internet Protocol (TCP/IP) was created.
- As various network systems migrated to this standard, they could then communicate with any network using the protocol. The Internet was born.



Key Milestones in Internet History

- 1969: Launch of ARPANET, the first operational packet-switching network.
- 1971: The first email is sent by Ray Tomlinson.
- 1983: Adoption of TCP/IP as the standard protocol for ARPANET.
- 1989–1990: Tim Berners-Lee invents the World Wide Web.
- 1993: The first graphical web browser, Mosaic, is released.
- 1998: Google is founded, revolutionizing search engines.
- 2007: The iPhone introduces widespread mobile Internet access.
- 2010s: Rise of cloud computing, social media, and streaming services.



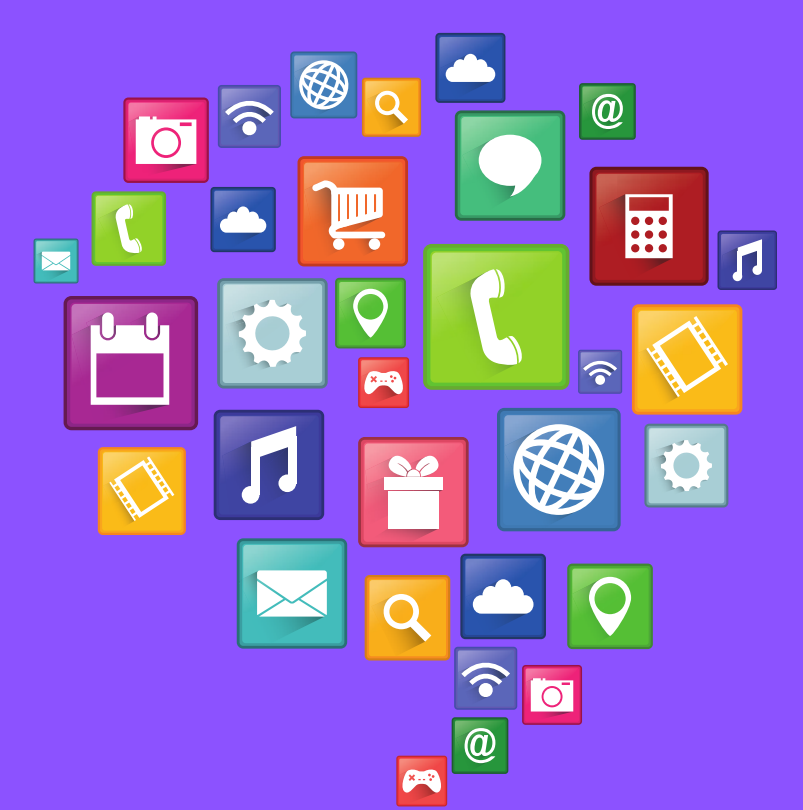
Who Invented the Internet?

- While no single person can claim to have “invented” the Internet, it emerged through the collaborative efforts of numerous brilliant minds over several decades.

Vint Cerf and Bob Kahn Known as “The Fathers of the Internet”

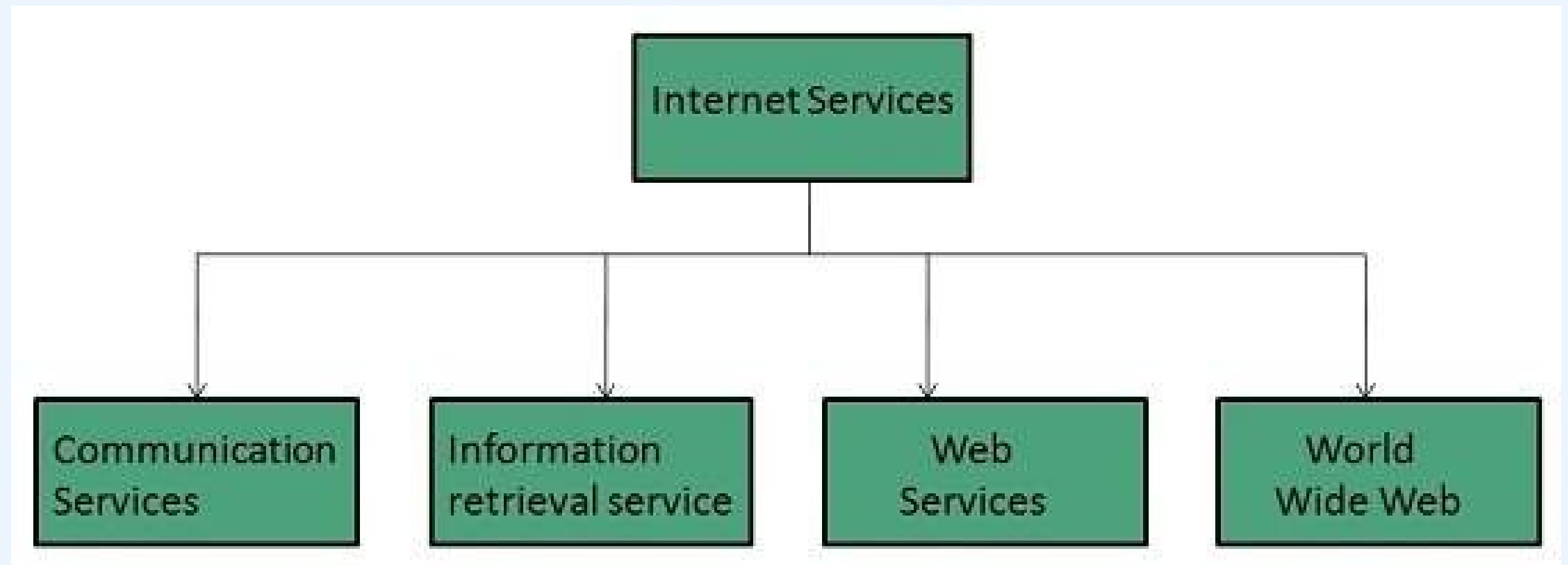
Key Contributions:

- Invented TCP/IP protocol in 1974
- Created the foundation for data transmission
- Developed internet architecture
- Established basic internet communication standards



Services Internet

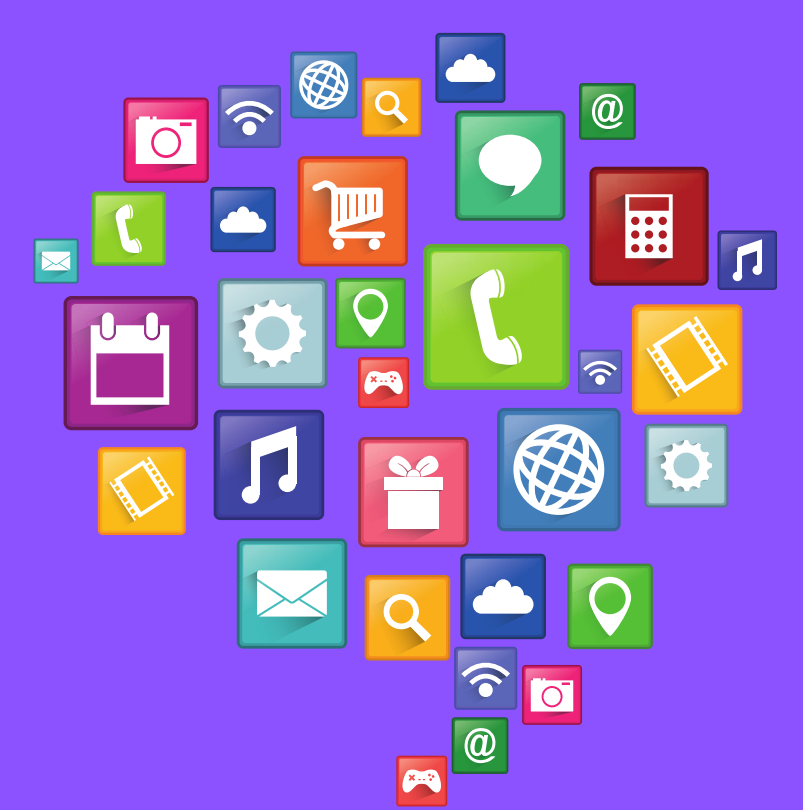
Internet Services allows us to access huge amount of information such as text, graphics, sound and software over the internet. Following diagram shows the four different categories of Internet Services.



Communication Services

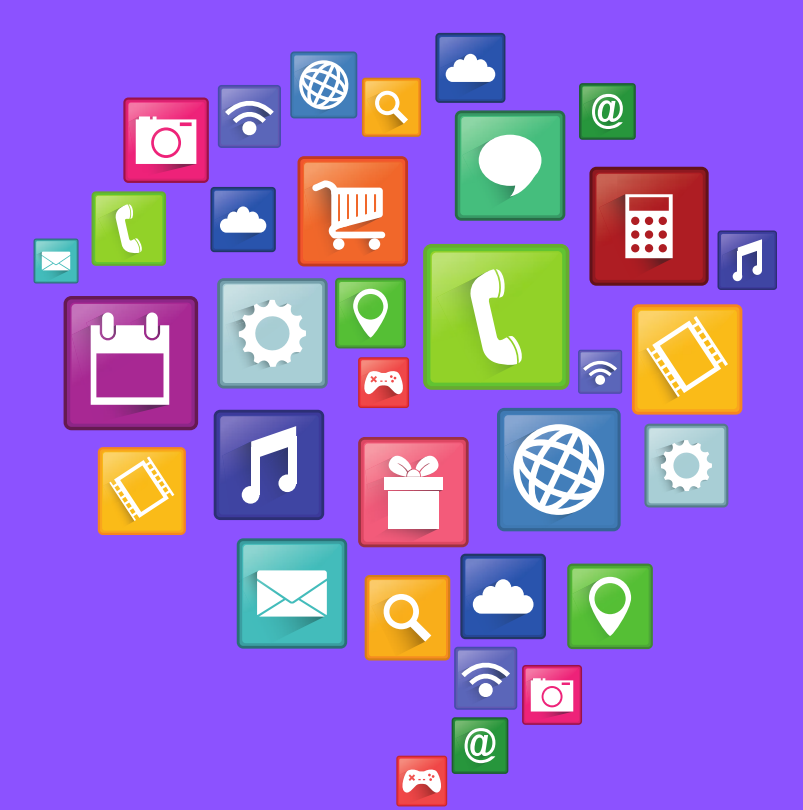
There are various Communication Services available that offer exchange of information with individuals or groups.

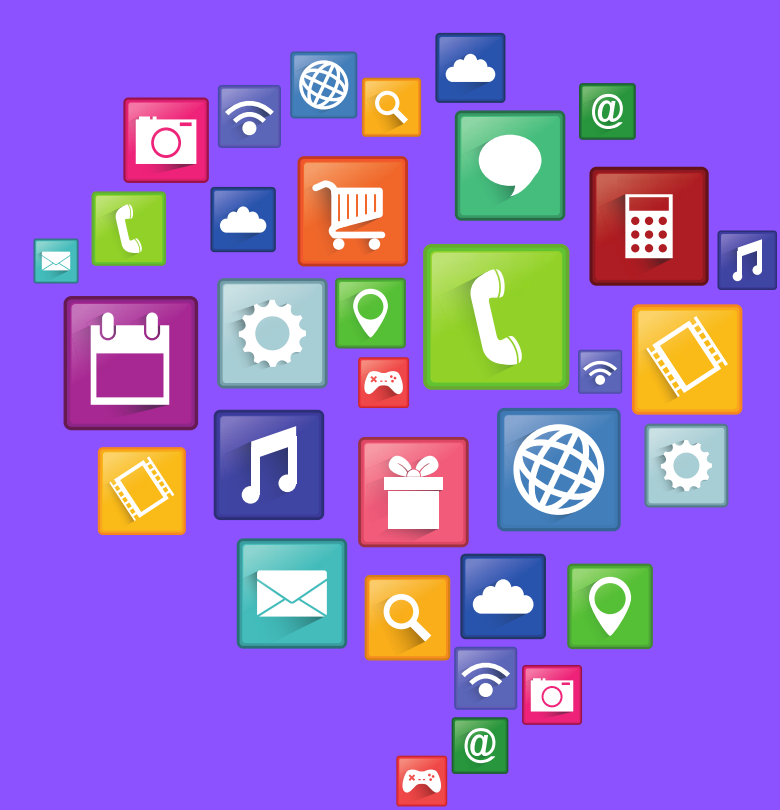
- Electronic Mail: Used to send electronic message over the internet.
- Telnet: Used to log on to a remote computer that is attached to internet.
- Newsgroup: Offers a forum for people to discuss topics of common interests.
- Internet Relay Chat (IRC): Allows the people from all over the world to communicate in real time.



Communication Services

- Mailing Lists: Used to organize group of internet users to share common information through e-mail.
- Internet Telephony (VoIP): Allows the internet users to talk across internet to any PC equipped to receive the call.
- Instant Messaging: Offers real time chat between individuals and group of people. Eg. Yahoo messenger, MSN messenger.





Information Retrieval Services

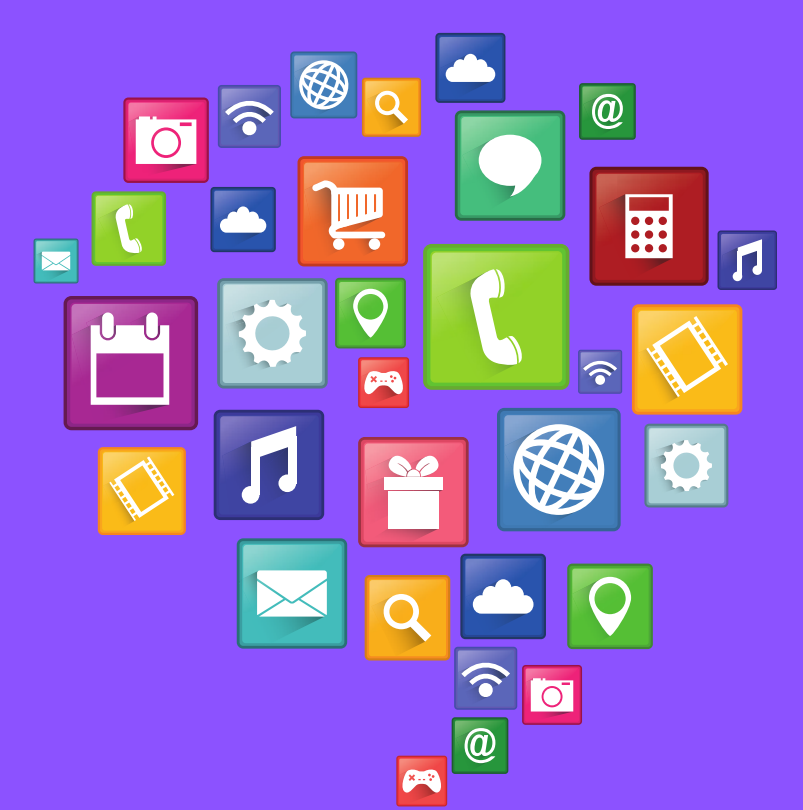
- File Transfer Protocol (FTP): Enable the users to transfer files.

Web Services

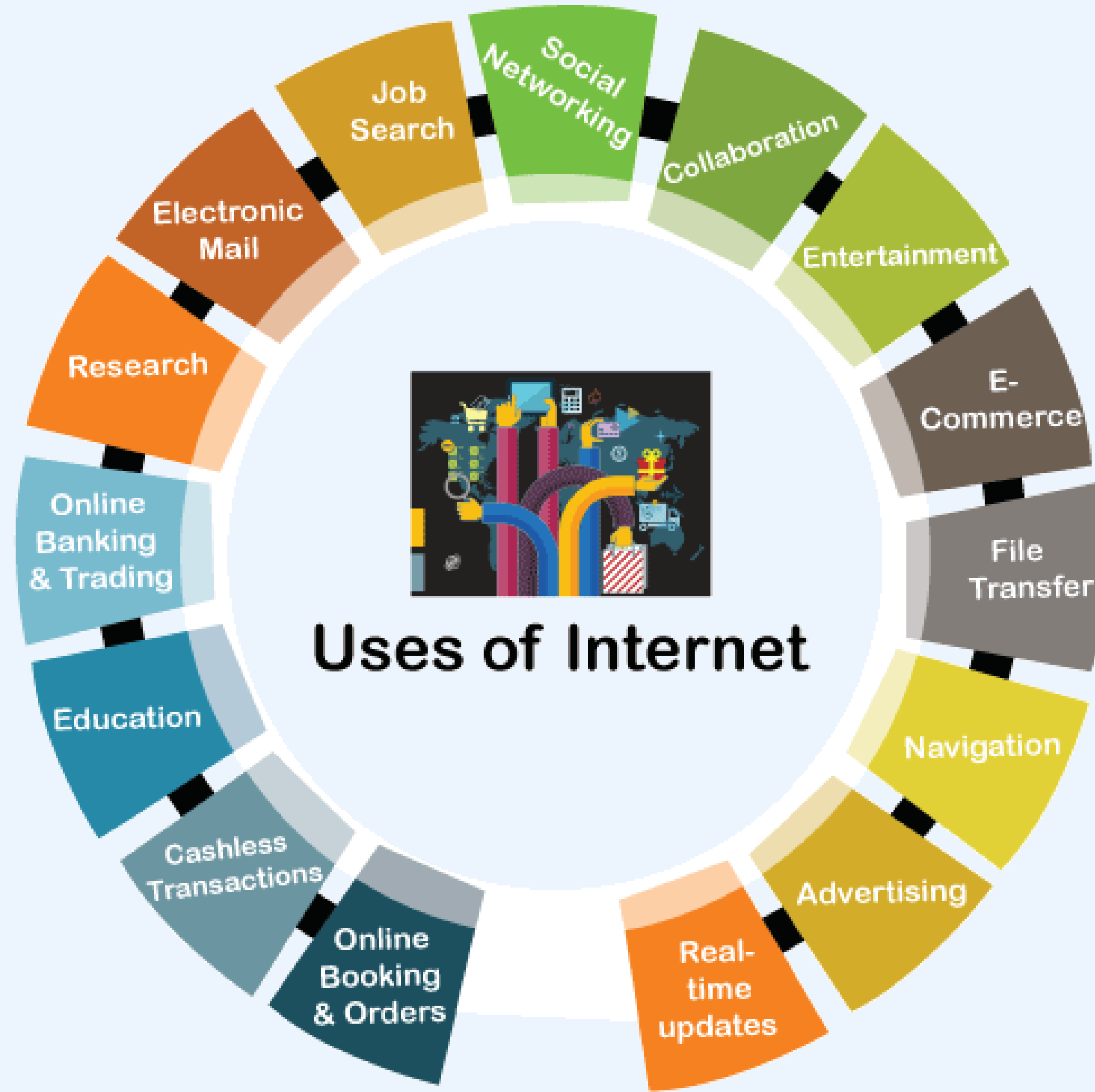
- Web services allow exchange of information between applications on the web. Using web services, applications can easily interact with each other.

World Wide Web (WWW)

- WWW is also known as W3. It offers a way to access documents spread over the several servers over the internet. These documents may contain texts, graphics, audio, video, hyperlinks. The hyperlinks allow the users to navigate between the documents.



Uses of Internet



Hypertext and its significance



- Hypertext most often refers to text on a computer that will lead the user to other, related information on demand.
- that is the ability, in a computer environment, to move from one part of a document to another or from one document to another through internal connections among these documents (called "hyperlinks");
- Hypertext allows system designers to organize information in a branching structure instead of a linear one.

World Wide Web



- The World Wide Web and its associated Hyper Text Transfer Protocol (HTTP) grew out of work done at the European laboratory for particle physics (CERN) in 1990
- Tim Berners-Lee developed HTTP as a network protocol for distributing documents and wrote the first web browser.
- HTTP is simple request/response protocol in which web browser asks for a document and the web server return the document in the form of HTML data stream preceding by a few descriptive headers.

World Wide Web

- Also called web.
- Is the most popular application or service of an internet.
- A collection of interlinked multimedia documents that are stored on the Internet and accessed using a common protocol (HTTP)
- Web pages could be static (fixed) or dynamic (changing).
- A Web site is a collection of related Web pages stored on the same web server.
- A Web Server is a computer that delivers requested Web pages to your computer. The same Web server can store multiple Web sites.
- WWW is just a collections of websites around the world.

Introduction to web programming



- Web programming involves creating web pages, web applications, and other online content that can be displayed in a web browser.
- Web programming is used to create websites, online stores, web applications, and other online services that are accessed through the Internet
- Web programming is accomplished using a variety of programming languages, including HTML, CSS, JavaScript, PHP, Python, Ruby, and Java.

Introduction to web programming

Web Development can be classified into two ways:

- **Frontend Development**
- **Backend Development**

Frontend Development

- The part of a website where the user interacts directly is termed as front end. It is also referred to as the 'client side' of the application.

Backend Development

- Backend is the server side of a website. It is part of the website that users cannot see and interact with. It is the portion of software that does not come in direct contact with the users. It is used to store and arrange data.



Introduction to web programming

Key Components of Web Programming

- **Client-Side Technologies:** Client-side technologies are web technologies that run on a user's device, usually within the user's browser. such as HTML, CSS, and JavaScript, enabling dynamic user interfaces and interactive web.
- **Server-Side Technologies:** A form of web server technology in which users' requests are fulfilled by running a script directly on the web server to generate dynamic HTML pages.
- It is used to provide interactive web sites capable of interfacing with databases and other data stores.
- Server-Side Technologies include: **Server-Side Languages, Databases and Server Environment.**



Introduction to web programming

Client-Side Technologies

- **HTML:** stands for HyperText Markup Language. It is used to design web pages using a markup language.
- HTML is the combination of **Hypertext** and **Markup language**.
- Hypertext defines the link between the web pages.
- A markup language is used to define the text document within tag which defines the structure of web pages.



Introduction to web programming

Client-Side Technologies



- **CSS** : Cascading Style Sheets is a stylesheet language used to design the webpage to make it attractive.
- A stylesheet language is used to describe the presentation and formatting of a document written in a markup language such as HTML.
- The reason for using CSS is to simplify the process of making web pages presentable.
- CSS allows you to apply styles to web pages.
- More importantly, CSS enables you to do this independent of the HTML that makes up each web page.

Introduction to web programming

Client-Side Technologies

- **JavaScript:** JavaScript is the most powerful and versatile scripting language used in the web.
- It is a lightweight, cross-platform, single-threaded and interpreted scripting language.

Lightweight: lightweight programming language, means that it can run on almost any device without requiring a lot of memory or processing power, is easy to implement.

Single-threaded: refers to a program design where tasks are executed sequentially

cross-platform: designed to function on multiple operating systems or environments without requiring significant modifications

- It is a commonly used programming language to create dynamic and interactive elements in web applications. It is easy to learn.



Introduction to web programming

Server-Side Technologies

- **Server-Side Languages:** Languages like PHP, Python, Ruby, Java, and Node.js are used to handle server-side logic, process requests, and generate dynamic content.
- **Databases:** Systems like **MySQL, PostgreSQL, MongoDB** are used to store and retrieve data dynamically, enabling web applications to manage and manipulate information.
- **Server Environment:** Software environments such as Apache, Microsoft IIS (Internet Information Services) provide the infrastructure to host and serve web applications. They handle incoming requests, route them to the appropriate handlers, and send responses back to clients.



Introduction to web programming

How the Web Work?

The stream of event that occurs with every web page that appears on screen.



1. You request a web page by either typing its URL(for e.g `http://www.wsu.edu.et`) directly in the browser or by clicking on a link on the page. The URL contains all the information needed to target a specific document on a specific web server on the internet.
2. your browser send an HTTP request on the server named in the URL and asks for a specific file .if the URL specifies a directory(not a file).it is the same as request the default file in that directory.

Introduction to web programming

How the Web Work?

3. The server looks for the request file and sends an either of the following HTTP response:

If the page cannot be found. The server return an error message . the message typical says “404” not found.

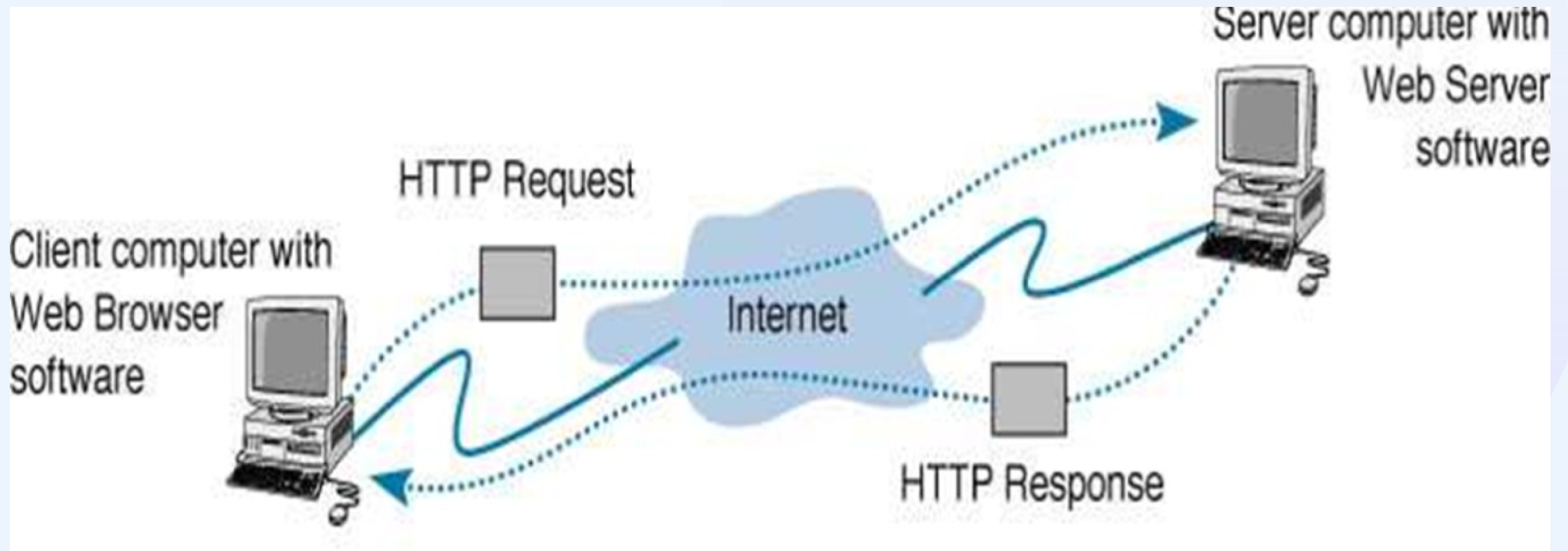
- If the document is found. The server retrieves the requested file and returns it to the browser.

4. The browser parses the HTML document. If the page contains images(indicated by the HTML img tag), the browser contacts the server again to request each image files specified in the markup

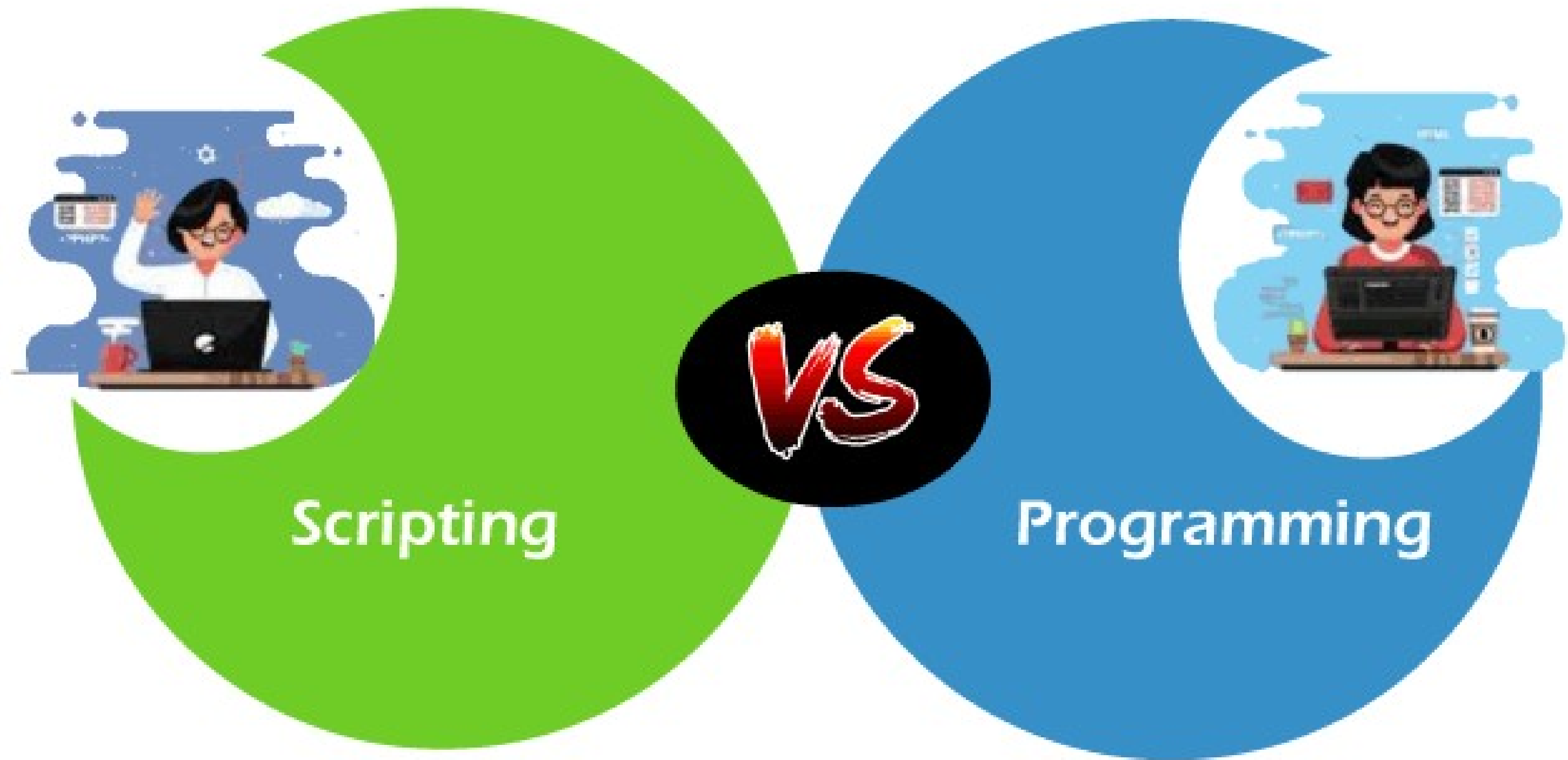
5. The browser inserts each image in the document flow where indicated by the img. Then, the assembled web page is displayed for your viewing pleasure.



How the Web Work?



Programming vs. Scripting



Programming vs. Scripting

- Both scripting and programming are computer languages.
- The basic difference between a scripting language and a programming language is that scripting languages do not need an additional step of compilation and rather they are **interpreted**, whereas programming languages are compiled and hence need a **compilation** step to convert the high-level language to machine code.
- **Programming Languages:** are compiled languages, which means the source code is translated (compiled) into machine code **once** and then executed by the hardware

Examples of pure compiled languages are C, C++, Erlang, Haskell, Rust, and Go.

Programming vs. Scripting

- **Interpreted Languages:** Interpreters run through a program line by line and execute each command (The code is compiled and executed each time the program is run).
- Interpreted languages were once significantly slower than compiled languages. But, with the development of just-in-time compilation, that gap is shrinking.
- **Just-in-time compilation:** is a method for improving the performance of interpreted programs. During execution the program may be compiled into native code to improve its performance. It is also known as dynamic compilation.

Examples of common interpreted languages are PHP, Ruby, Python, and JavaScript.

Programming vs. Scripting

- "A scripting language is a type of programming language which does not require explicit compilation step, and it is designed for a runtime system to automate the execution of tasks."

For example, a JavaScript program is not needed to be compiled before we run it. These are also known as very high-level programming languages because of working at a high level of abstraction.

- Scripting languages support "script," which is small program written for a specific runtime environment. These are interpreted at runtime rather than compiled. It means, to convert the source code to machine code, scripting languages use an **interpreter**, not the compiler. As scripting language is **not compiled** so as we write something meaningful, we can run it immediately.

Programming vs. Scripting

Advantages of scripting language

- It is an easy and quick process to learn coding in Scripting language
- In scripting languages, a wide variety of libraries is available that enable the developers to develop new applications.
- With the help of scripting languages, we can add visualization interfaces and combinations to web pages. Most of the latest web pages need scripting languages for creating enhanced web pages, fascinating UI, and many more.
- There are less number of data structures and variable to be used, which make it highly efficient.
- These are Less code-intensive as compared to traditional programming languages.



Commonly used scripting languages in web development

- Client-side and server-side scripting are two essential components of web programming that enable the creation of dynamic websites.

The server-side scripting and client-side scripting collaborate with one another to create a dynamic websites. However, these two scripting techniques are different, where:

- the client-side scripting focuses on developing the web application or website's interface to be more attractive and functional.
- On the other hand, server-side scripting mainly focuses on data access techniques, quick processing, error handling, etc.

Client-side scripting



- Client-side scripting refers to code that runs on the user's browser (executed on the client end without needing server-side processing).
- Client-side scripting may be utilized to check the user's form for problems before submitting it and to change the content based on the user input.
- The web needs three components to function client, database, and server.
- The client-side scripting may significantly reduce server demand.

Client-side scripting

- **Example** : JavaScript is a client-side scripting language designed for a specific purpose, but several JavaScript frameworks are already utilized as server-side scripting technologies.



Client-side scripting

Some main features of the client-side scripting are:

- It is intended to execute code on which a web browser runs, and the results of the inputs are delivered to an accessible user.
- Client-side scripting enables greater involvement with clients via the browser and is used to validate programs and functionality based on the request.
- The client does not include any contact with the server in client-side scripting; the only interaction is receiving the requested data.



server-side scripting

- server-side scripting involves code that executes on the web server. In other words, server-side scripting is any scripting method that may operate on a web server.
- At the server end, actions such as website customization, dynamic changes in website content, response creation to user requests, database access, and many more are carried out.
- There are various server-side scripting languages. Some main server-side scripting languages are: Python, PHP, Ruby etc.

server-side scripting

Some main features of the server-side scripting are:

- It is connected with data access, error handling, and data processing speed.
- It is processed and interacts with the server.
- Using a highly integrated programming language makes it more secure than client-side scripting.

Client-side scripting Vs server-side scripting

Client-side scripting

- client-side scripting is utilized at the front end, which users may access via the browser and the source code is visible to the user.
- Its main function is to provide the requested output to the end user.
- It reduce the server load.
- It runs on the user's computer.
- It doesn't offer any access to the files on the web servers.
- It is less secure than server-side scripting due to the code accessibility offered to the client.

server-side scripting

- Server-side scripting is utilized in the backend when the source code is invisible or hidden on the client side. output of server-side is an HTML page
- Its primary function is to manipulate and provide access to the respective database as per the request.
- It may effectively customize web pages and offer dynamic websites.
- It runs on the webserver.
- It offers complete access to the file that is stored in the web database server.
- It is more secure.

Client-Server Model

- The Client-server model is a distributed application structure that partitions tasks or workloads between the **providers of a resource or service**, called **servers**, and **service requesters** called **clients**.
- In the client-server architecture, when the client computer sends a request for data to the server through the internet, the server accepts the requested process and delivers the data packets requested back to the client.
- **Clients** are devices or programs that request services or resources, while
- **The server** is a powerful machine providing these resources or services.

Client-Server Model

- This architecture allows efficient data management and resource sharing, making it popular in web applications, databases, and other network-based systems.
- By separating roles and distributing tasks, client-server architecture enhances performance, scalability, and security.

Client-Server Architecture

